

Supporting information for

Tracking Two Decades of Global Gallium Stocks and Flows: A Dynamic

Material Flow Analysis

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S1 Abbreviated letter index table

Table S1. List of abbreviations

Abbreviation	Full name
PV	Photovoltaic panel
ZEV	Zero-emission vehicle
CIGS	Copper-indium-gallium-selenium thin-film solar cell
LED	Light-emitting diode
ICEV	Internal combustion engine vehicle
IC	Integrated circuit
MOCVD	Metal Organic Chemical Vapor Deposition
MBE	Molecular beam epitaxy
TMG	Trimethylgallium
EoL	End-of-life

S2 Calculating methods for each process of Ga cycle

S2.1 Ga in Aluminium cycle

Mass balances in Aluminium cycle were calculated by the following equations.

$$I_{Bayer}(t) = P_{bauxite}(t) * (1 - Loss_{bauxite}(t)) * M_{bauxite}(t) * (1 - T_O(t)) * (1 - Loss_{red}(t)) * (1 - Loss_{Al}(t)) \quad (S1)$$

$$I_{Bayer}(t) = Loss_{Bayer}(t) + I_{Al-Ga}(t) \quad (S2)$$

Where $I_{Bayer}(t)$ stands for Ga inflow in Bayer liquid in year t, $P_{bauxite}(t)$ stands for bauxite production in year t, $Loss_{bauxite}(t)$ stands for bauxite loss rate in the beneficiation process in year t, $M_{bauxite}(t)$ stand for Ga fraction in bauxite in year t, $T_O(t)$ stands for proportion of other uses of bauxite in year t, $Loss_{red}(t)$ stand for fraction of Ga lost to red mud in year t, $Loss_{Al}(t)$ stand for fraction of Ga lost to Aluminum metal in year t, $Loss_{Bayer}(t)$ stand for fraction of Ga lost to Bayer liquid in year t, $I_{Al-Ga}(t)$ stands for the amount of Ga from Bayer liquid to Ga production stage in year t.

S2.2 Ga in Zinc cycle

Mass balances in zinc cycle were calculated by the following equations.

$$I_{t-slag}(t) = P_{ore}(t) * (1 - Loss_{ore}(t)) * M_{ore}(t) * (1 - Loss_{metal}(t)) \quad (S3)$$

$$I_{t-slag}(t) = Loss_{slag}(t) + I_{Zn-Ga}(t) \quad (S4)$$

Where $I_{t-slag}(t)$ stands for total Ga inflow in zinc slag in year t, $P_{ore}(t)$ stands for zinc ore production in year t, $Loss_{ore}(t)$ stands for zinc ore loss rate in the beneficiation process in year t, $M_{ore}(t)$ stand for Ga fraction in zinc ore in year t, $Loss_{metal}(t)$ stand for fraction of Ga lost to zinc metal in year t, $Loss_{slag}(t)$ stand for fraction of Ga lost to zinc slag in year t, $I_{Zn-Ga}(t)$ stands for the amount of Ga from zinc slag to Ga production stage in year t.

S2.3 Ga production

Mass balances in Ga production were calculated by the following equations.

$$I_{pri}(t) = I_{Al-Ga}(t) * (1 - R_{Al-Ga}(t)) + I_{Zn-Ga}(t) * (1 - R_{Zn-Ga}(t)) \quad (S5)$$

$$I_{tpro}(t) = I_{pri}(t) + I_{sec}(t) \quad (S6)$$

$$I_{sec}(t) = (T_{sem}(t) + T_{eol}(t)) * (1 - R_{sec}(t)) \quad (S7)$$

$$I_{tpro}(t) = O_{pro}(t) \quad (S8)$$

Where $I_{pri}(t)$ stands for the primary Ga production in year t, $R_{Al-Ga}(t)$ stands for loss rate of Ga extracted from Baier liquid, $R_{Zn-Ga}(t)$ stands for loss rate of Ga extracted from zinc slag, $I_{sec}(t)$ stands for secondary Ga production in year t, $T_{sem}(t)$ and $T_{eol}(t)$ stand for the amount of Ga new and old scrap collected in year t respectively, $R_{sec}(t)$ stands for loss rate in the smelting stage of Ga scrap, $I_{tpro}(t)$ stands for total production of Ga, $O_{pro}(t)$ stands for Ga outflow in Ga production stage.

S2.4 Production of semi-products

Mass balances in production of semi-products were calculated by the following equations.

$$I_{sem}(t) = O_{pro}(t) \quad (\text{S9})$$

$$I_{fab}(t) * MS_i(t) * (1 - Loss_i(t)) = O_i(t) \quad (\text{S10})$$

$$\sum_i I_{fab}(t) * MS_i(t) * R_i(t) = T_{sem}(t) \quad (\text{S11})$$

Where $I_{sem}(t)$ stands for Ga inflow in semi-products making stage in year t, $MS_i(t)$ stands for market share of Ga inflow in semi-product j in year t, $Loss_i(t)$ stands for the Ga loss rate in semi-product i making stage in year t, $O_i(t)$ stands for Ga flow out in semi-product i making stage in year t, $R_i(t)$ stands for the Ga recycling rate in semi-product j making stage in year t.

S2.5 Production of final products

Mass balances in production of final products were calculated by the following equations.

$$I_j(t) = \sum_i O_i(t) \quad (\text{S12})$$

Where $I_j(t)$ stands for the Ga inflow in final product j in year t.

S2.6 In-use stock & outflow

Mass balances in in-use stocks & outflows were calculated by the following equations.

$$O_j(t) = \int_{t_0}^t I_j(t) * pdf(t, t') dt' \quad (\text{S13})$$

$$S_j(t) = \int_{t_0}^t (I_j(t) - O_j(t)) dt \quad (\text{S14})$$

Where $pdf(t, t')$ is the life distribution function (Weibull distribution), and t , is the age of the finished product from production to year t , $O_j(t)$ for Ga outflow in the finished product j , $S_j(t)$ for Ga in-use stock in finished product j .

For Weibull distribution

$$pdf(t, t') = \int_0^t (1 - \exp(-(\frac{t-t'}{\alpha})^\beta)) dt' \quad (\text{S15})$$

For each product, lifetime distribution and parameters are available in Table S1.

Table S2. Lifetime distributions and parameters for finished products

Applications	Distribution function	Scale parameter α	Shape parameter β	Source
LED lightings	Weibull distribution	10	2	(Løvik et al., 2016; Song et al., 2022)
Mobile phones		3.28	2.32	(He et al., 2018)
Computers		8.4	2.2	(Althaf et al., 2021; Charpentier Poncelet et al., 2022; Song et al., 2022)
Wind turbines		20	4	(Løvik et al., 2016)
ICEVs		16	4.8	(Løvik et al., 2016; Wu et al., 2020)
ZEVs		16	4.8	(Løvik et al., 2016; Wu et al., 2020)
Solar PVs		25	5	(Løvik et al., 2016)
Large durables		6	2	(Song et al., 2022; Wang et al., 2022)
Others		6	2	(Song et al., 2022)

S2.7 Waste management

$$I_{re}(t) = O_{re}(t) = \sum_j O_j(t) * (1 - DR(t)) \quad (\text{S18})$$

Where $I_{re}(t)$ and $O_{re}(t)$ stand for the Ga inflow and outflow in waste management stage (recycling process) in year t . $DR(t)$ is the dissipation ratio of finished product j in year t .

S3 Complete overview of the parameters and their respective

sources

Table S3. Main parameters and primary data of global Ga cycle

Process	Description	Unit	Detailed data value	Source	uncertainty range
Gallium in Aluminum cycle	Historic Bauxite production	t	Time series	(USGS)	Low
	Ga mass fraction in bauxite	(t/t)	5.4e-005	(Løvik et al., 2016)	Medium
	Fraction of Ga lost to red mud	1	0.38	(Løvik et al., 2015)	Medium
	Fraction of Ga lost to alumina	1	0.6	(Løvik et al., 2015)	Medium
	The proportion of alumina produced by the Bayer liquor	1	0.85	(Li et al., 2021; Løvik et al., 2015)	Medium
	The proportion of bauxite used for others (not including the production of alumina by Bayer liquor)	1	0.15	(Li et al., 2021; Løvik et al., 2015)	Medium
Gallium in Zinc cycle	Zinc ore production	t	Time series	(USGS, n.d.)	Low
	Gallium mass fraction in Zinc ore	(t/t)	5.0e-005	(Frenzel et al., 2016; Jansson et al., 2016)	Medium
	Fraction of Ga lost to zinc	1	0.02	(Frenzel et al., 2016)	Medium
	Mass fraction of Ga in zinc refinery residue	1	0.0035	(Rao et al., 2019)	Medium
	Fraction of Ga lost to zinc refinery residue	1	0.98	(Frenzel et al., 2016)	Medium
Gallium production	Rate of Ga extraction from zinc slag	1	0.9	(Frenzel et al., 2016)	Medium
	Rate of gallium extraction from Bayer liquor	1	0.95	(Løvik et al., 2015)	Medium
	Yield in recycling of wafer production scrap	1	0.92	(Løvik et al., 2016, 2015)	Medium
Semi-products manufacturing	Market share of Ga inflow in semi-products	1	Time series	(Løvik et al., 2016; Shanghai Metals Market (SMM), 2022)	Low
	GaN-based LED market share	1	0.7	Own assumption	High
	GaAs/GaP-based LED market share	1	0.3	(Løvik et al., 2016)	Medium

Yield in production of sputtering targets/MBE sources	1	1	(Løvik et al., 2016)	Medium
Yield in production of TMG	1	0.92	(Løvik et al., 2016)	Medium
Yield in production of GaAs/GaP substrates	1	0.35	(Løvik et al., 2016)	Medium
Collection rate GaAs/GaP substrate production scrap	1	0.63	(Løvik et al., 2016)	Medium
Deposition yield MOCVD	1	0.11	(Løvik et al., 2016)	Medium
Share of GaAs/GaP LED fabricated on GaAs	1	0.88	(Løvik et al., 2016)	Medium
Loss rate due to dicing of LEDs	1	0.31	(Løvik et al., 2016)	Medium
Loss rate due to non-working devices LED	1	0.08	(Løvik et al., 2016)	Medium
Collection rate of back-grinding/epitaxial lift-off scrap LEDs	1	0.3	(Løvik et al., 2016)	Medium
Collection rate other LED fab. Scrap	1	0	(Løvik et al., 2016)	Medium
Concentration of Ga in GaN	g/cm ³	5.12	(Løvik et al., 2016)	Medium
Concentration of Ga in GaAs	g/cm ³	2.56	(Løvik et al., 2016)	Medium
Concentration of Ga in GaP	g/cm ³	2.87	(Løvik et al., 2016)	Medium
Gallium intensity in AlInGaP	g/cm ³	0.96	(Løvik et al., 2016)	Medium
Thickness of GaAs substrate	m	6.53*10 ⁽⁻⁴⁾	(Løvik et al., 2016)	Medium
Thickness of GaP substrate	m	5.62*10 ⁽⁻⁴⁾	(Løvik et al., 2016)	Medium
Thickness of deposition layer of LEDs	m	4*10 ⁽⁻⁶⁾	(Løvik et al., 2016)	Medium
Thickness of LED chips	m	1.4*10 ⁽⁻⁴⁾	(Løvik et al., 2016)	Medium
Thickness of IC chips	m	1*10 ⁽⁻⁴⁾	(Løvik et al., 2016)	Medium
Thickness of deposition layer	m	2*10 ⁽⁻⁶⁾	(Løvik et al., 2016)	Medium
Deposition yield MBE	1	0.07	(Løvik et al., 2016)	Medium
Share of integrated circuits produced with MOCVD	1	0.67	(Løvik et al., 2016)	Medium
Share of integrated circuits produced with MEB	1	0.33	(Løvik et al., 2016)	Medium

				2016)	
	Collection rate back-grinding scrap IC	1	0.1	(Løvik et al., 2016)	Medium
	Collection rate other scrap IC fab.	1	0	(Løvik et al., 2016)	Medium
	Loss rate due to dicing of ICs	1	0.15	(Løvik et al., 2016)	Medium
	Loss rate due to non-working devices ICs	1	0.08	(Løvik et al., 2016)	Medium
	Loss rate due to non-working photovoltaics	1	0.08	(Løvik et al., 2016)	Medium
	Deposition yield of CIGS	1	0.45	(Løvik et al., 2016)	Medium
	Collection rate non-working photovoltaics scrap	1	0.7	(Løvik et al., 2016)	Medium
	Collection rate other photovoltaics scrap	1	0	(Løvik et al., 2016)	Medium
	Yield in production of NdFeB magnets	1	0.7	(Liu et al., 2022; Løvik et al., 2016)	Medium
	Collection rate NdFeB magnets scrap	1	0	(Liu et al., 2022; Løvik et al., 2016)	Medium
Final products manufacturing	Market share Consumption structure of all kinds of semi-products	t	Time series	(Shanghai Metals Market (SMM), 2022)	Low
Waste management	Recycling rate of EoL final products	1	0	Own assumption	High
	Lifetime distributions and parameters for finished products	-	-	Table S1	Medium

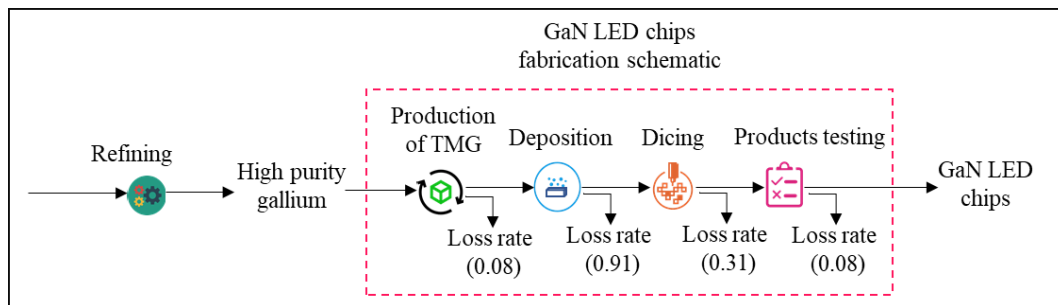


Fig. S1. Framework of GaN LED chips production process

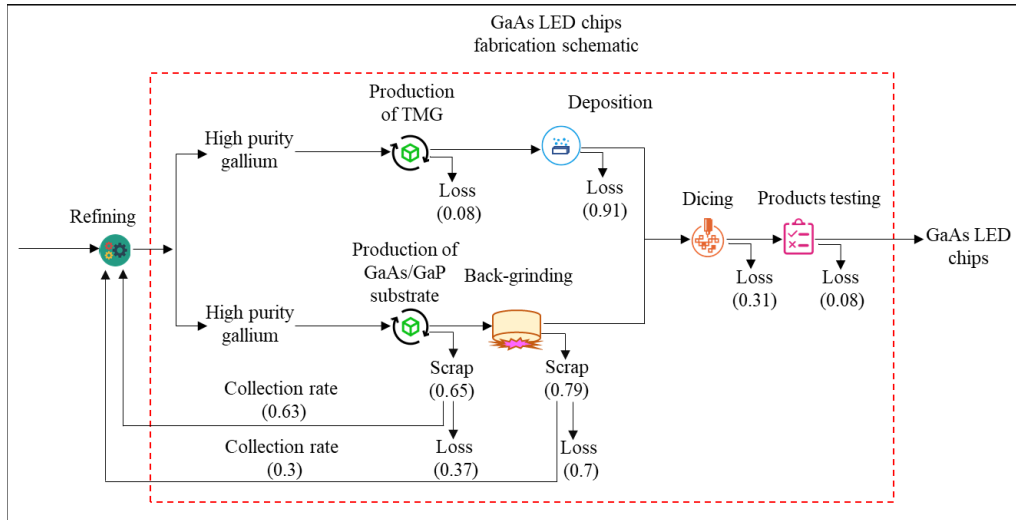


Fig. S2. Framework of GaAs LED chips production process

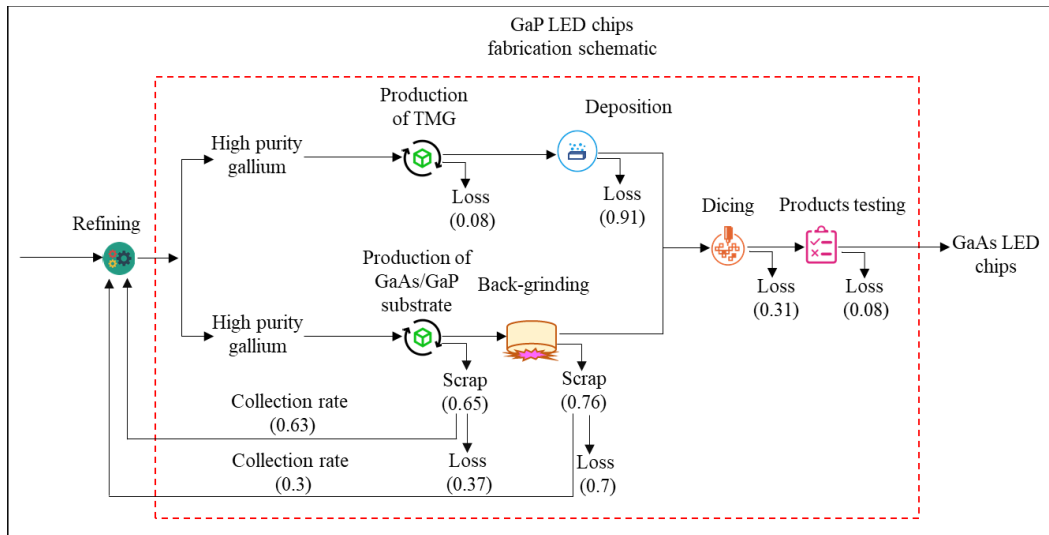


Fig. S3. Framework of GaP LED chips production process

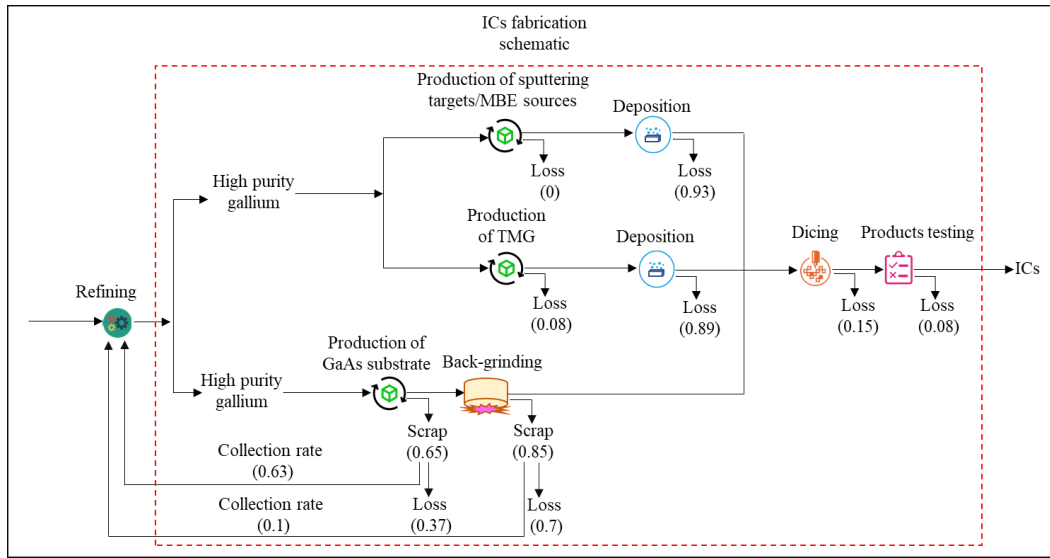


Fig. S4. Framework of ICs production process

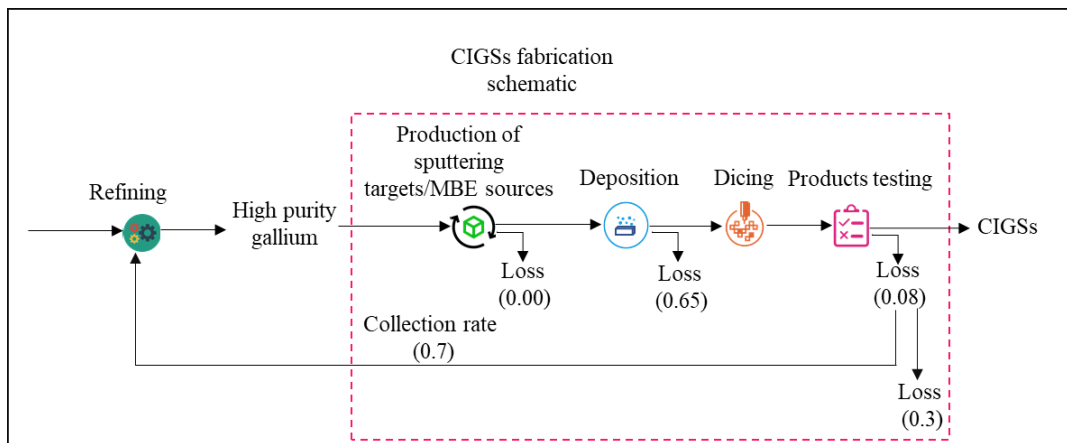


Fig. S5. Framework of CIGSs production process

Table S4. Ga consumption structure of semi-products

	GaN LEDs	GaAs/ GaP LEDs	ICs	CIGSs	NdFeB Magnets	Others
2000	7%	21%	28%	2%	26%	16%
2001	7%	19%	31%	1%	29%	13%
2002	7%	19%	31%	2%	30%	12%
2003	7%	21%	27%	2%	30%	13%
2004	7%	21%	25%	2%	31%	15%
2005	7%	20%	26%	3%	30%	14%
2006	6%	18%	31%	2%	27%	15%
2007	7%	18%	32%	2%	29%	12%
2008	7%	21%	36%	2%	22%	12%
2009	8%	23%	35%	2%	19%	13%
2010	8%	23%	35%	2%	19%	13%
2011	8%	23%	29%	2%	22%	16%
2012	8%	23%	32%	2%	23%	12%
2013	9%	24%	33%	2%	21%	11%
2014	9%	24%	37%	2%	20%	9%
2015	8%	24%	34%	2%	19%	13%
2016	8%	23%	35%	2%	19%	12%
2017	8%	23%	38%	2%	19%	9%
2018	8%	22%	38%	2%	20%	10%
2019	8%	22%	40%	2%	20%	8%
2020	7%	21%	42%	2%	19%	9%
2021	7%	21%	44%	2%	19%	7%

Table S5. The proportion of GaN LEDs flow to final products

	LED lightings	Mobile phones	Computers	Wind turbines	ICEVs	ZEVs	Solar PVs	Large durables
2000	22.0%	8.0%	20.0%	0.0%	2.0%	0	0.0%	48.0%
2001	22.0%	8.0%	20.0%	0.0%	2.0%	0	0.0%	48.0%
2002	22.0%	8.0%	20.0%	0.0%	2.0%	0	0.0%	48.0%
2003	22.0%	8.0%	20.0%	0.0%	2.0%	0	0.0%	48.0%
2004	22.0%	8.0%	20.0%	0.0%	2.0%	0	0.0%	48.0%
2005	22.0%	8.0%	20.0%	0.0%	2.0%	0	0.0%	48.0%
2006	22.0%	8.0%	20.0%	0.0%	2.0%	0	0.0%	48.0%
2007	22.0%	8.0%	20.0%	0.0%	2.0%	0	0.0%	48.0%
2008	22.0%	8.0%	20.0%	0.0%	2.0%	0	0.0%	48.0%
2009	22.0%	8.0%	20.0%	0.0%	2.0%	0	0.0%	48.0%
2010	22.0%	8.0%	20.0%	0.0%	2.0%	0	0.0%	48.0%
2011	22.0%	8.0%	20.0%	0.0%	2.0%	0	0.0%	48.0%
2012	30.0%	9.0%	19.0%	0.0%	2.3%	0	0.0%	39.6%
2013	38.0%	10.0%	18.0%	0.0%	2.6%	0	0.0%	31.2%
2014	41.0%	10.5%	17.0%	0.0%	2.9%	0	0.0%	28.3%
2015	44.0%	11.0%	16.0%	0.0%	2.9%	0	0.0%	25.4%
2016	47.0%	11.5%	15.0%	0.0%	3.4%	0	0.0%	22.5%
2017	48.0%	12.0%	14.0%	0.0%	3.4%	0	0.0%	21.6%
2018	49.0%	12.5%	13.0%	0.0%	3.1%	0	0.0%	20.7%
2019	50.0%	13.0%	12.0%	0.0%	3.2%	0	0.0%	19.8%
2020	51.0%	13.5%	11.0%	0.0%	2.6%	0	0.0%	18.9%
2021	52.0%	14.0%	10.0%	0.0%	2.1%	0	0.0%	18.0%

Table S6. The proportion of GaAs/GaP LEDs flow to final products

	LED lightings	Mobile phones	Computers	Wind turbines	ICEVs	ZEVs	Solar PVs	Large durables
2000	17.0%	3.0%	5.0%	0	12.0%	0.0%	0	63.0%
2001	17.0%	3.0%	5.0%	0	12.0%	0.0%	0	63.0%
2002	17.0%	3.0%	5.0%	0	12.0%	0.0%	0	63.0%
2003	17.0%	3.0%	5.0%	0	12.0%	0.0%	0	63.0%
2004	17.0%	3.0%	5.0%	0	12.0%	0.0%	0	63.0%
2005	17.0%	3.0%	5.0%	0	12.0%	0.0%	0	63.0%
2006	17.0%	3.0%	5.0%	0	12.0%	0.0%	0	63.0%
2007	17.0%	3.0%	5.0%	0	12.0%	0.0%	0	63.0%
2008	17.0%	3.0%	5.0%	0	12.0%	0.0%	0	63.0%
2009	17.0%	3.0%	5.0%	0	12.0%	0.0%	0	63.0%
2010	17.0%	3.0%	5.0%	0	11.9%	0.1%	0	63.0%
2011	17.0%	3.0%	5.0%	0	11.8%	0.2%	0	63.0%
2012	20.0%	4.0%	5.5%	0	25.2%	0.8%	0	44.5%
2013	24.0%	5.0%	6.0%	0	23.6%	1.4%	0	40.0%
2014	28.0%	6.0%	6.5%	0	21.7%	2.3%	0	35.5%
2015	32.0%	7.0%	7.0%	0	18.6%	4.4%	0	31.0%
2016	36.0%	8.0%	7.5%	0	18.7%	3.3%	0	26.5%
2017	40.0%	9.0%	8.0%	0	16.2%	4.8%	0	22.0%
2018	44.0%	10.0%	8.5%	0	13.0%	7.0%	0	17.5%
2019	48.0%	11.0%	9.0%	0	11.7%	7.3%	0	13.0%
2020	52.0%	12.0%	9.5%	0	8.3%	9.7%	0	8.5%
2021	55.0%	13.0%	10.0%	0	5.9%	11.1%	0	5.0%

Table S7. The proportion of ICs flow to final products

	LED lightings	Mobile phones	Computers	Wind turbines	ICEVs	ZEVs	Solar PVs	Large durables
2000	1.0%	5.0%	55.0%	0	5.0%	0.0%	0	34.0%
2001	1.0%	6.0%	54.0%	0	5.2%	0.0%	0	33.8%
2002	1.0%	7.0%	53.0%	0	5.5%	0.0%	0	33.5%
2003	1.0%	8.0%	52.0%	0	5.7%	0.0%	0	33.3%
2004	1.0%	9.0%	51.0%	0	6.0%	0.0%	0	33.0%
2005	1.0%	10.0%	50.0%	0	6.2%	0.0%	0	32.8%
2006	1.0%	11.0%	49.8%	0	6.5%	0.0%	0	31.8%
2007	1.0%	12.0%	49.5%	0	6.7%	0.0%	0	30.8%
2008	1.0%	13.0%	49.3%	0	7.0%	0.0%	0	29.8%
2009	1.0%	14.0%	49.0%	0	7.2%	0.0%	0	28.8%
2010	2.0%	15.0%	48.5%	0	7.4%	0.1%	0	27.0%
2011	3.0%	17.0%	48.0%	0	7.6%	0.1%	0	24.3%
2012	4.0%	20.0%	46.5%	0	7.7%	0.3%	0	21.5%
2013	5.0%	22.0%	45.0%	0	7.5%	0.5%	0	20.0%
2014	6.0%	24.0%	43.5%	0	7.2%	0.8%	0	18.5%
2015	7.0%	26.0%	42.0%	0	6.5%	1.5%	0	17.0%
2016	8.0%	28.0%	40.5%	0	6.8%	1.2%	0	15.5%
2017	9.0%	30.0%	39.0%	0	6.2%	1.8%	0	14.0%
2018	10.0%	32.0%	37.5%	0	5.2%	2.8%	0	12.5%
2019	11.0%	34.0%	36.0%	0	4.9%	3.1%	0	11.0%
2020	12.0%	36.0%	34.5%	0	3.7%	4.3%	0	9.5%
2021	13.0%	38.0%	33.0%	0	2.8%	5.2%	0	8.0%

Table S8. The proportion of CIGSs flow to final products

	LED lightings	Mobile phones	Computers	Wind turbines	ICEVs	ZEVs	Solar PVs	Large durables
2000							100%	
2001							100%	
2002							100%	
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2016							100%	
2017							100%	
2018							100%	
2019							100%	
2020							100%	
2021							100%	

Table S9. The proportion of NdFeB magnets flow to final products

	LED lightings	Mobile phones	Computers	Wind turbines	ICEVs	ZEVs	Solar PVs	Large durables
2000	0	1.2%	66.6%	9.5%	5.1%	0.0%	0	17.6%
2001	0	5.1%	69.0%	5.9%	3.9%	0.0%	0	16.1%
2002	0	6.7%	67.7%	4.5%	6.0%	0.0%	0	15.0%
2003	0	5.4%	71.0%	2.6%	5.3%	0.0%	0	15.7%
2004	0	4.6%	75.1%	2.2%	4.7%	0.0%	0	13.5%
2005	0	5.2%	75.0%	2.4%	5.6%	0.0%	0	11.8%
2006	0	7.5%	68.4%	3.4%	7.1%	0.0%	0	13.6%
2007	0	9.4%	59.2%	6.5%	10.2%	0.0%	0	14.7%
2008	0	9.0%	53.9%	11.6%	10.8%	0.0%	0	14.7%
2009	0	13.5%	40.8%	18.3%	10.1%	0.0%	0	17.3%
2010	0	14.0%	38.7%	19.1%	8.0%	0.1%	0	20.0%
2011	0	16.0%	24.7%	28.8%	11.8%	0.2%	0	18.6%
2012	0	14.0%	34.8%	23.8%	9.9%	0.3%	0	17.2%
2013	0	17.9%	26.0%	24.7%	10.8%	0.7%	0	20.0%
2014	0	17.8%	23.3%	25.8%	10.7%	1.2%	0	21.3%
2015	0	15.4%	21.2%	32.0%	9.3%	2.2%	0	19.9%
2016	0	14.6%	14.6%	35.2%	11.9%	2.1%	0	21.6%
2017	0	11.3%	12.7%	35.4%	11.8%	3.5%	0	25.3%
2018	0	9.3%	9.1%	35.0%	11.4%	6.1%	0	28.9%
2019	0	8.8%	10.1%	36.5%	10.2%	6.4%	0	28.0%
2020	0	6.3%	10.8%	40.4%	7.0%	8.1%	0	27.4%
2021	0	5.7%	10.3%	38.7%	6.6%	12.2%	0	26.5%

S4 Ga loss during the manufacturing of semi-products

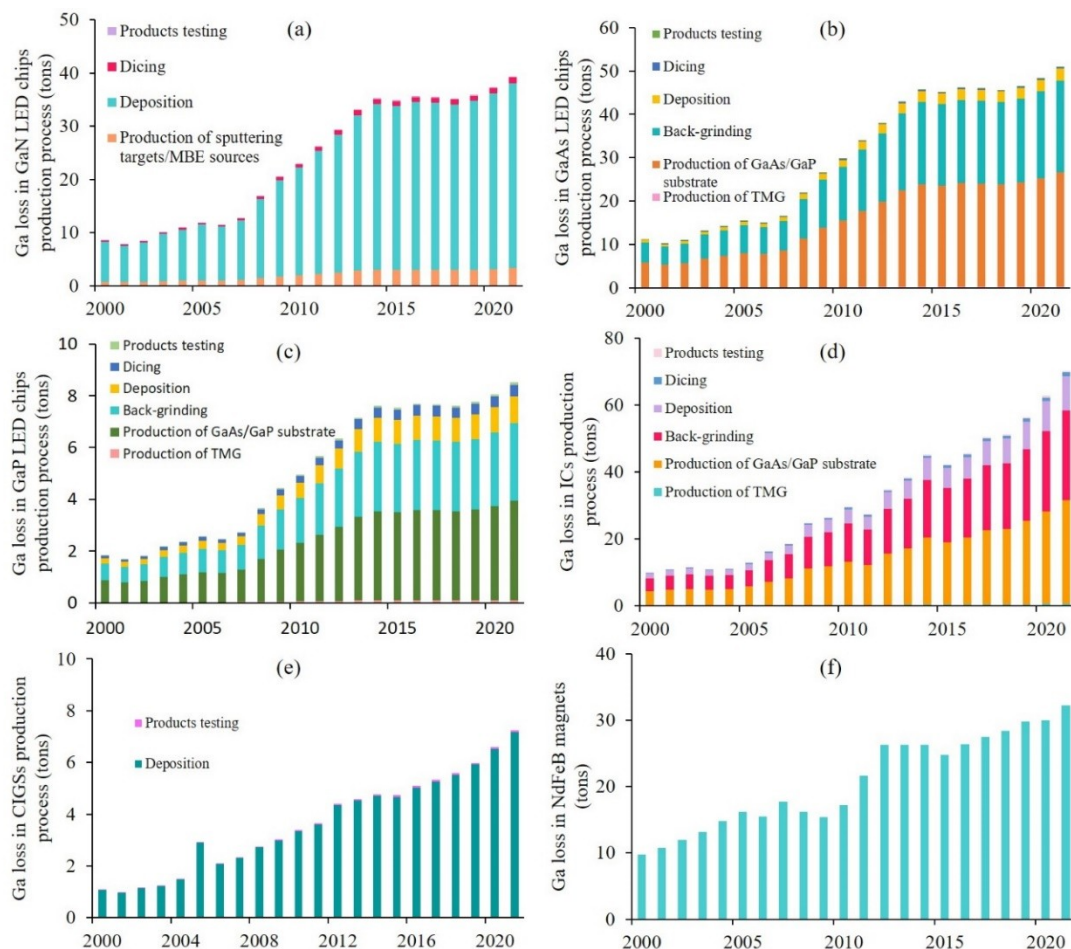
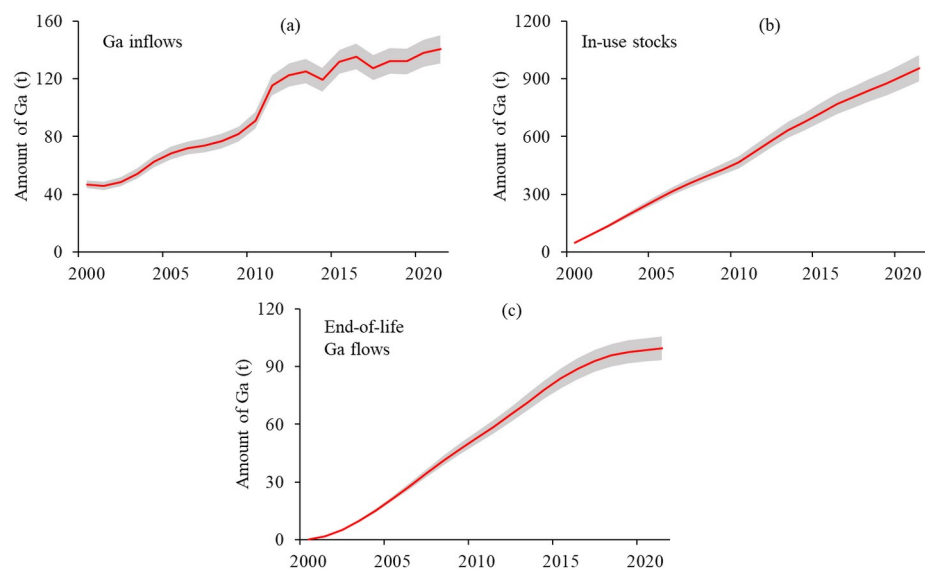


Fig. S6. Historical trends in Ga loss during the manufacturing of semi-products. Ga loss in GaN LED chips (a), Ga loss in GaAs LED chips (b), Ga loss in GaP LED chips (c), Ga loss in ICs (d), Ga loss in CIGSs (e) and Ga loss in NdFeB magnets. TMG = Trimethylgallium, MEB = Molecular beam epitaxy.

S5 Uncertainty analysis results of global Ga flows and stocks

Fig. S7. Uncertainty analysis results of global Ga inflows (a), in-use stocks (b) and end-of



life Ga flows (c).

S6 Ga loss information in Semi-product manufacturing stage and strategies for improving he circularity

Table S10. Ga loss information in Semi-product manufacturing stage from 2000 to 2021 and strategies for improving he circularity

Semi-products	Subsegment	Cumulative Ga loss (tons)	Reasons	Improving the circularity of Ga
GaN LEDs production stage	Production of TMG	43.9	Lack of effective recycling measures.	Enhancing recovery of a variety of intermediates, e.g., gallium trichloride, magnesium-based gallium iodide, and ether complexes of trimethyl gallium. Exploring techniques to improve deposition efficiency. In metal organic chemical vapor deposition (MOCVD), improvements in reactor design, precursor materials, and process control should be ongoing to enhance deposition efficiency and yield.
	Deposition	459.5	Low deposition efficiency.	
	Dicing	14.09	-	-
	Products testing	2.5	-	-
GaAs/GaP LEDs production stage	Production of TMG	4.6	-	-
	Production of GaAs/GaP substrate	381.6	Low deposition efficiency.	Optimizing parameters such as temperature, gas flow, and pressure in the vapor deposition chamber is essential to ensure the uniform growth of gallium arsenide and enhance the yield.
	Back-grinding	305.5	1. High dicing loss rate before the backgrinding. 2. Low scrap collection rate.	1. Optimizing the technical process parameters for various grinding processes like rotary grinding, double-sided grinding, and other similar processes can significantly reduce material losses and improve yield in semiconductor manufacturing. 2. Invest in efficient waste collection systems that capture as much of the cut material as possible.
	Deposition	48.4	Low deposition efficiency	The measures to improve the cycle are the same as in the GaN LEDS production stage.
	Dicing	38.3	Low economic efficiency of recycling.	Utilize efficient dissolution and leaching processes to selectively extract Ga from the GaAs/GaP LED scraps.
	Products testing	6.8	-	
	Production of TMG	6.5	-	
ICs	Production of GaAs/GaP substrate	311.7	Low deposition efficiency.	Optimizing parameters such as temperature, gas flow, and pressure in the vapor deposition chamber is essential to ensure the uniform growth of gallium arsenide and enhance the yield.
	Back-grinding	269.9	1. High dicing loss rate before the backgrinding. 2. Low scrap collection rate.	The measures to improve the cycle are the same as in the GaAs/GaP LEDs production stage.
	Deposition	101.5	Low deposition efficiency.	The measures to improve the cycle are the same as in the GaN LEDS production stage.

CIGSs	Dicing	11.8	-	
	Products testing	5.4	-	
	Deposition	79.5	Low deposition efficiency.	The measures to improve the cycle are the same as in the GaN LEDS production stage.
	Products testing	1.0	-	-
NdFeB magnets	-	458.5	Low economic efficiency of recycling.	Utilize efficient dissolution and leaching processes to selectively extract gallium from the NdFeB magnet scraps.

S7 World Alumina Refinery Production and amount of Ga in Bayer liquor

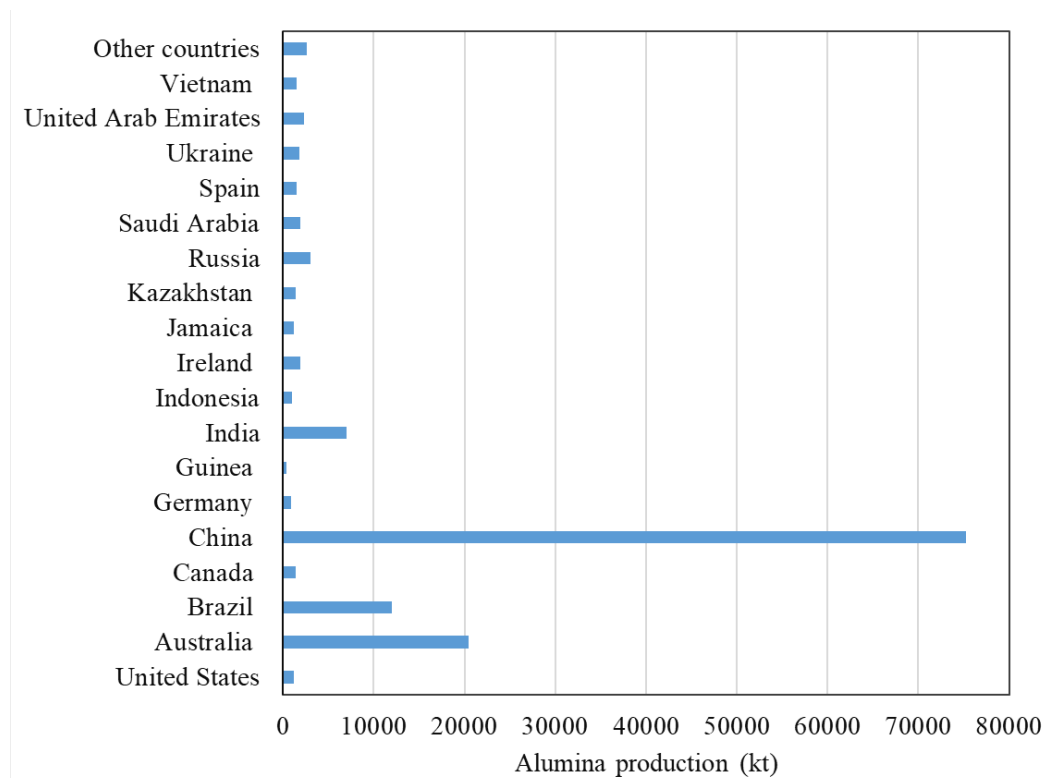


Fig. S8. World Alumina Refinery Production

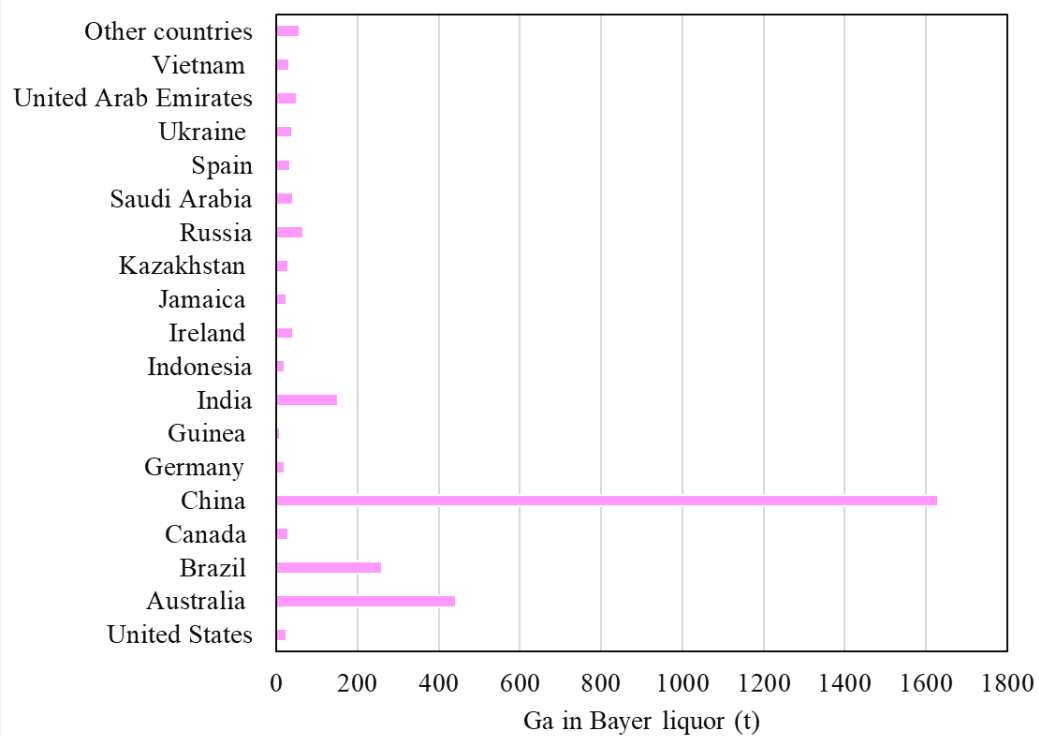


Fig. S9. Amount of Ga in Baier liquid in countries in 2021

S8 The standard techniques for production Ga from Bayer liquid

The standard techniques for production Ga from Bayer liquid include solvent extraction, ion exchange, electrochemistry, and adsorption (Kou et al., 2022; Roosen et al., 2017). The electrochemical method has a simple process and equipment. Still, the highly toxic cathode electrolysis of mercury is seriously harmful to the environment and human health and has been banned by most countries (Chai et al., 2023). The extractant used in solvent extraction method is expensive, and the extractant dissolved in the seed mother liquor also has adverse effects on the electrolysis in the post-sequence process (Lu et al., 2017). The ion exchange process is simple, high recovery, and easy to operate, but it is limited by vanadium and aluminum co-extraction and resin degradation (Lu et al., 2017). Currently, adsorption stands out as the most cost-effective method for Ga extraction, offering a combination of low expenses, high efficiency, operational adaptability, suitability for low concentrations, and environmentally sustainable practices (Chai et al., 2023; Lu et al., 2017; Roosen et al., 2017).

References

- Althaf, S., Babbitt, C.W., Chen, R., 2021. The evolution of consumer electronic waste in the United States. *Journal of Industrial Ecology* 25, 693–706. <https://doi.org/10.1111/jiec.13074>
- Chai, N., Gao, L., Li, S., Cao, Y., Ma, Z., Li, L., Hu, M., 2023. In-suit ion-imprinted bio-sorbent with superior adsorption performance for gallium(III) capture. *Journal of Cleaner Production* 387, 135861. <https://doi.org/10.1016/j.jclepro.2023.135861>
- Charpentier Poncelet, A., Helbig, C., Loubet, P., Beylot, A., Muller, S., Villeneuve, J., Laratte, B., Thorenz, A., Tuma, A., Sonnemann, G., 2022. Losses and lifetimes of metals in the economy. *Nature Sustainability* 5, 717–726. <https://doi.org/10.1038/s41893-022-00895-8>
- Frenzel, M., Ketris, M.P., Seifert, T., Gutzmer, J., 2016. On the current and future availability of gallium. *Resources Policy* 47, 38–50. <https://doi.org/10.1016/j.resourpol.2015.11.005>
- He, P., Wang, C., Zuo, L., 2018. The present and future availability of high-tech minerals in waste mobile phones: Evidence from China. *Journal of Cleaner Production* 192, 940–949. <https://doi.org/10.1016/j.jclepro.2018.04.222>
- Jansson, N.F., Malmström, L., Zetterqvist, A., Allen, R.L., 2016. A comment on the occurrence of gallium and germanium in the Zinkgruvan Zn-Pb-Ag-(Cu) sulphide deposit, Bergslagen, Sweden. *Gff* 138, 533–535. <https://doi.org/10.1080/11035897.2016.1165284>
- Kou, Z., Bian, W., Wang, C., 2022. Preparation of novel amidoxime resin and its performance in gallium (III) extraction from Bayer liquor. *Chemical Engineering Journal Advances* 11, 100317. <https://doi.org/10.1016/j.cej.2022.100317>
- Li, Q., Dai, T., Gao, T., Zhong, W., Wen, B., Li, T., Zhou, Y., 2021. Aluminum material flow analysis for production, consumption, and trade in China from 2008 to 2017. *Journal of Cleaner Production* 296, 126444. <https://doi.org/10.1016/j.jclepro.2021.126444>
- Liu, Q., Sun, K., Ouyang, X., Sen, B., Liu, L., Dai, T., Liu, G., 2022. Tracking Three Decades of Global Neodymium Stocks and Flows with a Trade-Linked Multiregional Material Flow Analysis. *Environmental Science & Technology*. <https://doi.org/10.1021/acs.est.2c02247>
- Løvik, A.N., Restrepo, E., Müller, D.B., 2016. Byproduct Metal Availability Constrained by Dynamics of Carrier Metal Cycle: The Gallium-Aluminum Example. *Environmental Science and Technology* 50, 8453–8461. <https://doi.org/10.1021/acs.est.6b02396>
- Løvik, A.N., Restrepo, E., Müller, D.B., 2015. The global anthropogenic gallium system: Determinants

- of demand, supply and efficiency improvements. *Environmental Science and Technology* 49, 5704–5712. <https://doi.org/10.1021/acs.est.5b00320>
- Lu, F., Xiao, T., Lin, J., Ning, Z., Long, Q., Xiao, L., Huang, F., Wang, W., Xiao, Q., Lan, X., Chen, H., 2017. Resources and extraction of gallium: A review. *Hydrometallurgy* 174, 105–115. <https://doi.org/10.1016/j.hydromet.2017.10.010>
- Rao, S., Wang, D., Liu, Z., Zhang, K., Cao, H., Tao, J., 2019. Selective extraction of zinc, gallium, and germanium from zinc refinery residue using two stage acid and alkaline leaching. *Hydrometallurgy* 183, 38–44. <https://doi.org/10.1016/j.hydromet.2018.11.007>
- Roosen, J., Mullens, S., Binnemans, K., 2017. Chemical immobilization of 8-hydroxyquinoline and 8-hydroxyquinoline on chitosan-silica adsorbent materials for the selective recovery of gallium from Bayer liquor. *Hydrometallurgy* 171, 275–284. <https://doi.org/10.1016/j.hydromet.2017.05.026>
- Shanghai Metals Market (SMM), 2022. Market analysis of gallium industry chain in China.
- Song, H., Wang, C., Sen, B., Liu, G., 2022. China Factor: Exploring the Byproduct and Host Metal Dynamics for Gallium-Aluminum in a Global Green Transition. *Environmental Science and Technology* 56, 2699–2708. <https://doi.org/10.1021/acs.est.1c04784>
- USGS, n.d. Bauxite and Alumina Statistics and Information [WWW Document]. URL <https://www.usgs.gov/centers/national-minerals-information-center/bauxite-and-alumina-statistics-and-information> (accessed 7.3.23a).
- USGS, n.d. Zinc Statistics and Information [WWW Document]. URL <https://www.usgs.gov/centers/national-minerals-information-center/zinc-statistics-and-information> (accessed 7.3.23b).
- Wang, Q.C., Chen, W.Q., Wang, P., Dai, T., 2022. Illustrating the supply chain of dysprosium in China through material flow analysis. *Resources, Conservation and Recycling* 184, 106417. <https://doi.org/10.1016/j.resconrec.2022.106417>
- Wu, Y., Yang, L., Tian, X., Li, Y., Zuo, T., 2020. Temporal and spatial analysis for end-of-life power batteries from electric vehicles in China. *Resources, Conservation and Recycling* 155, 104651. <https://doi.org/10.1016/j.resconrec.2019.104651>